

Hu 0825

a)

	1	2	3	4	5	6	7
p	8.9	8.1	9.3	7.7	10.4	8.3	7.4
1	8.9	8.1	9.3	7.7	10.4	8.3	7.4
2	8.9	7.4	9.0	7.8	9.4	8.1	6.9
12	0.1	0.7	0.3	1.1	0.5	0.2	0.5
rank	1.5	1.7	4	1.5	8.5	3	5.5
signed	1.5	1.7	4	1.5	8.5	3	5.5

H₀: 2 有 有 有 有

H_a: 2 有 有 有 有

检验统计量 = 0.05

$$\frac{0}{1} = 0.025$$

$$T^+ = 1.5 + 7 + 4 + 5.5 + 3 + 5.5$$

$$= 26.5$$

$$T^- = 1.5$$

$$\min(T^+, T^-) = T^- = 1.5$$

$$T_{7, 0.025}(\chi^2_{12}) = 2$$

$\alpha = 0.05$

$$T^- < 2, \text{ so reject } H_0$$

b) p-value = $P(X \leq 1) + P(X \geq 6)$

$$= [P(1) + P(2) + P(3) + P(4) + P(5) + P(6) + P(7) + P(8)]_{0.125}$$

$$= 0.125 \times 8 = 1.0$$

not same, because sign test only consider sign, not value.

11.

Month	A	B	12	signed rank	signed rank
1	3	7	4	6	(+)
2	14	12	2	3.5	(+)
3	7	9	2	3.5	(-)
4	10	15	5	7	(-)
5	9	12	3	5	(-)
6	6	6	0	—	—
7	13	12	1	1.5	(+)
8	6	5	1	1.5	(+)
9	7	13	6	8	(-)

$$T^+ = 3.5 + 1.5 + 1.5 = 6.5$$

$$T^- = 6 + 3.5 + 7 + 5 + 8$$

$$= 29.5$$

$$\min(T^+, T^-) = T^+ = 6.5$$

$$T_{9, 0.01} = 4$$

$$T^+ > T_{9, 0.01}, \text{ non-reject } H_0$$

1) 那为向 每月 breakdown rate 全不一

可能呈季节性, 环境, 材料等

0825-1

15) student	with large	without large	id	rank	sign	$T^- = 0$
1	20	5	15	17	(+)	$T^+ = \frac{20(21)}{2}$
2	24	9	15	17	(+)	$= 210$
3	20	5	15	17	(+)	$T_{20,0.05} = 52$
4	19	9	9	8.5	(+)	$M_h(T^+, T) = T = 0$
5	22	6	16	19	(+)	$T^* < 52, \text{ reject } H_0$
6	19	11	8	6.5	(+)	
7	20	8	12	13.5	(+)	
8	19	11	8	6.5	(+)	
9	17	7	10	11	(+)	() same result
10	21	9	12	13.5	(+)	sign to a
11	17	8	9	8.5	(+)	signed-rank test
12	20	16	4	2	(+)	
13	20	10	10	11	(+)	
14	16	12	4	2	(+)	
15	24	7	17	20	(+)	
16	22	9	13	15	(+)	
17	25	21	4	2	(+)	
18	21	14	7	4.5	(+)	
19	19	12	7	4.5	(+)	
20	23	13	10	11	(+)	

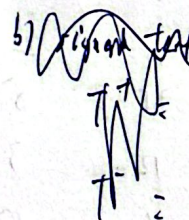
a) pair t-test

sign test

wilcoxon signed-rank test

the population must be Normal distribution.

No, because sample not enough big.



b) signed test

$T^+ \geq 20$

$H_0: p = 0.5$

$H_a: p \neq 0.5$

$$2P(X < 0) = 2[0.5^{20}] < 0.01$$

reject H_0

c. 4个 methods 有 20 个 数据

Training period

	1	1.5	2
0.5	4 (10)	4 (9)	4 (9)
5	7 (10)	6 (10)	7
1.1	5 (5)	7 (10)	5 (5)
9	8 (10)	8 (10)	5 (5)
12	9 (10)	6 (10)	4 (5)

0.5	1	1.5	2
8 (10)	9 (10)	4 (2)	4 (2)
(20) 14	7 (10)	6 (10)	7 (10)
(16) 9	5 (5)	7 (10)	5 (5)
(12) 12	8 (10)	8 (10)	5 (5)
(18) 10	9 (10)	6 (10)	4 (2)
86	60	39	24

Likelihood-ratio

H-test

$$H = \frac{12}{k(k+1)} \sum \frac{T_i^2}{n_i} - 3(k+1)$$

$$\sim \chi^2_{k-1, \alpha}$$

n=20

H: same distribution

H_a: different distribution

$$H = \frac{12}{4(5)} \left[\frac{86^2}{5} + \frac{60^2}{5} + \frac{39^2}{5} + \frac{24^2}{5} \right] - 3(5)$$

$$= 12.27$$

$\chi^2_{3, 0.01}$

$$\chi^2_{3, 0.01} = 11.345 < H = 12.27, \text{ so } \Rightarrow \text{reject } H_0$$

⊗ ⊗ ⊗ (这里公式中的 N 是 total 数) (这里 signed-rank 的 Rank test, n 为 total 个数)

8. 2.2 $R_1 = 247.5$ $R_2 = 168$ $R_3 = 216.5$ $R_4 = 188$, $N = 41$

$$H = \frac{12}{4(41)} \left[\frac{247.5^2}{10} + \frac{168^2}{10} + \frac{216.5^2}{10} + \frac{188^2}{10} \right] - 3(41)$$

$$= 2.632$$

$$\chi^2_{k-1, \alpha} = \chi^2_{3, 0.01} = 11.345 > H, \text{ non reject } H_0$$

ANOVA 的 p-value = 0.002

also reject H₀

b) p-value, $\chi^2_{3, 0.01} = 6.251$

$$\text{so } P(\chi^2_3 > 2.632) > P(\chi^2_3 > 6.251) = 0.001$$

10.

Analyze

Row	1	2	3	4
1	9.1(2)	2.2(1)	6.8(4)	6.2(3)
2	8.1(2)	9.2(3)	6.6(1)	9.6(4)
3	5.0(3)	2.6(1)	3.6(2)	6.5(4)
4	7.9(2)	9.4(4)	3.3(3)	9.5(1)
5	3.9(3)	7.4(4)	2.1(2)	2.0(1)

$$T_1 = 12 \quad T_2 = 13 \quad T_3 = 6 \quad T_4 = 15$$

$$F_t = \frac{12}{b(k+1)} \sum T_i^2 - 3b(k+1)$$

$$\frac{12}{20(5)} [12^2 + 13^2 + 6^2 + 15^2] - 3(5)(5)$$

$$= 1.56$$

$$\chi_{(b-1)(k-1), 0.05} = \chi_{4, 0.05} = 7.775 > 1.56 F_t$$

so we reject H_0

a) is Randomize block design

b) is Normal distribution

but σ^2 not

ANOVA $\bar{y}_{ij}$ is

6-89

irrelevant

(1)

↙

11.

	Location						Row Sum
v	1	2	3	4	5	6	
A	4	2	3	3	3	3	$R_A = 11$
B	1	3	2	2	2	2	$R_B = 12$
C	5	4	4	4	4	4	$R_C = 26$
D	3	5	5	5	5	5	$R_D = 27$
E	2	1	1	1	1	1	$R_E = 7$

H_0 : all distributions same

H_a : Not all same

a) use F_t , $\alpha = 0.05$

$$F_t = \frac{12}{b(k+1)} \sum T_i^2 - 3b(k+1)$$

$$= \frac{12}{30(6)} [12^2 + 12^2 + 26^2 + 27^2 + 7^2] - 3(6)(6)$$

$$= 20.33$$

$$\chi_{4, 0.05} = 9.488 < F_t, \text{ so reject } H_0$$

b) ANOVA test procedure $\alpha = 0.05$

so we reject H_0

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